

BIOS vs UEFI Comparison

Week 3 – ITEP 414 System Administration and Maintenance

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BSIT 4B

Category	BIOS	UEFI
Definition	Legacy firmware interface for hardware initialization and booting.	Modern firmware interface that replaces legacy BIOS.
Boot Process	POST → boot device → boot code	Firmware initialization → EFI boot manager/application → operating system boot loader.
Disk/Partition	Traditionally MBR-based with older limitations.	Commonly GPT-based and suitable for modern disks.
Security	Fewer modern pre-boot security features.	Supports Secure Boot on compatible systems.
Speed/Flexibility	Legacy-oriented.	More flexible for modern hardware and boot management.
Modern Usage	Legacy systems.	Modern PCs and servers.

Why UEFI Replaced BIOS

UEFI more closely aligns with the demands of contemporary computing. It supports GPT and other contemporary partitioning systems. enhanced boot management and has the potential to offer security features like Secure Boot. BIOS is still crucial for Although UEFI is the recommended firmware approach on contemporary hardware due to its support for legacy systems, current operating systems, storage, and security procedures more successfully